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Medication Tracking Application for Elderly Users

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Abstract

Medication adherence remains a critical challenge in healthcare, particularly among elderly populations who often manage multiple chronic conditions requiring complex medication schedules. Research indicates that approximately 50% of elderly patients do not take their medications as prescribed, leading to adverse health outcomes, increased hospitalizations, and higher healthcare costs estimated at \$100-300 billion annually in the United States alone.

This project develops a user-friendly medication tracking application designed specifically for elderly users and their caregivers. The application is cross-platform deployable as native desktop applications for Windows, macOS, and Linux, as well as a web application accessible through modern browsers.

Features

The system features comprehensive medication schedule management, intelligent reminder notifications, adherence tracking, caregiver monitoring capabilities, and secure local data storage. The interface emphasize accessibility with large fonts, high contrast ratios, simplified navigation, and large touch buttons suitable for elderly users with visual and motor impairments. Key functionalities include flexible medication scheduling supporting various intervals, quick medication entry, drug database integration, drug interaction warnings, medication quantity tracking with refill reminders, health measurement logging (blood pressure, glucose), and professional PDF report generation for healthcare provider consultations. The system supports multiple languages including Arabic and English with proper right-to-left (RTL) text support.

Objectives

Improve Adherence: Reduce missed doses through reliable, persistent reminders and simplified tracking.

Enhance Accessibility: Implement large fonts, high contrast interfaces, and intuitive navigation for seniors.

Empower Caregivers: Enable remote monitoring and instant missed dose alerts for peace of mind.

Privacy First: Prioritize local data storage (SQLite) to ensure sensitive health data remains private.

Cross-Platform: Native support for Windows, macOS, Linux, and Web from a single efficient codebase.

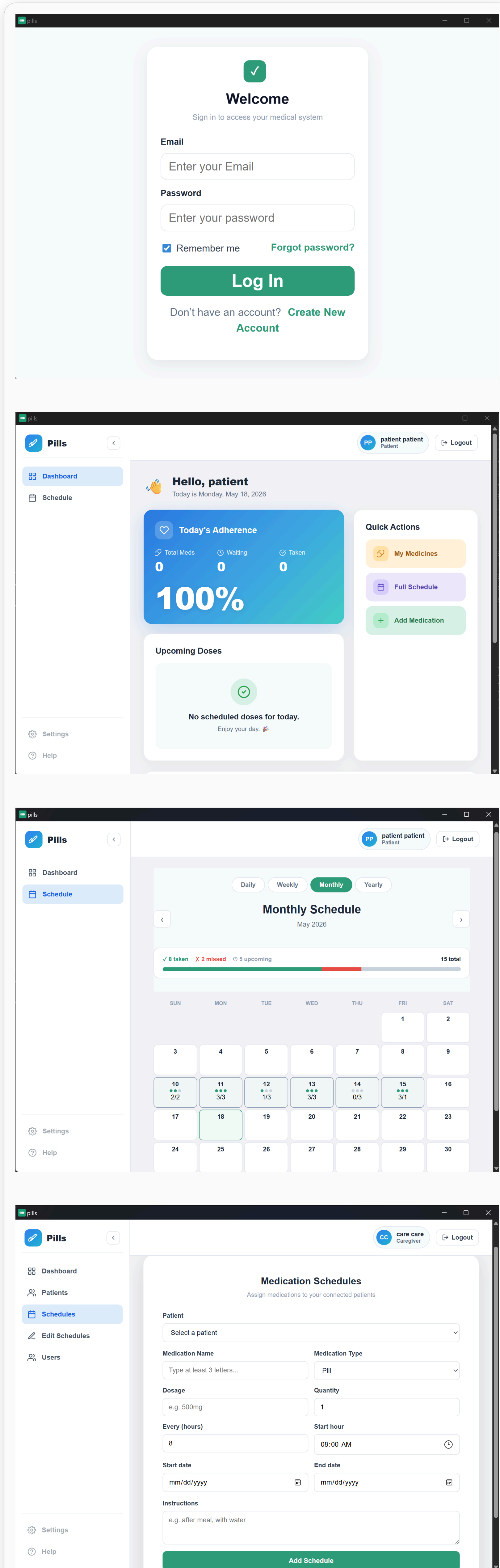
System Architecture

This project involves developing a user-friendly medication tracking application designed specifically for elderly users and their caregivers. The application will be built using Go programming language with the Wails framework to create cross-platform applications deployable as native desktop applications for Windows, macOS, and Linux, as well as a web application accessible through modern browsers. The technical architecture combines a high-performance Go backend for business logic and data management with a Svelte frontend built using modern web technologies (JavaScript, HTML, CSS), enabling dual deployment from a single codebase.

Conclusion

We designed and implemented a solution that bridges the gap between elderly patients and caregivers. By prioritizing privacy, accessibility, and maintaining a free cost model, we addressed some of the shortcomings of current solutions.

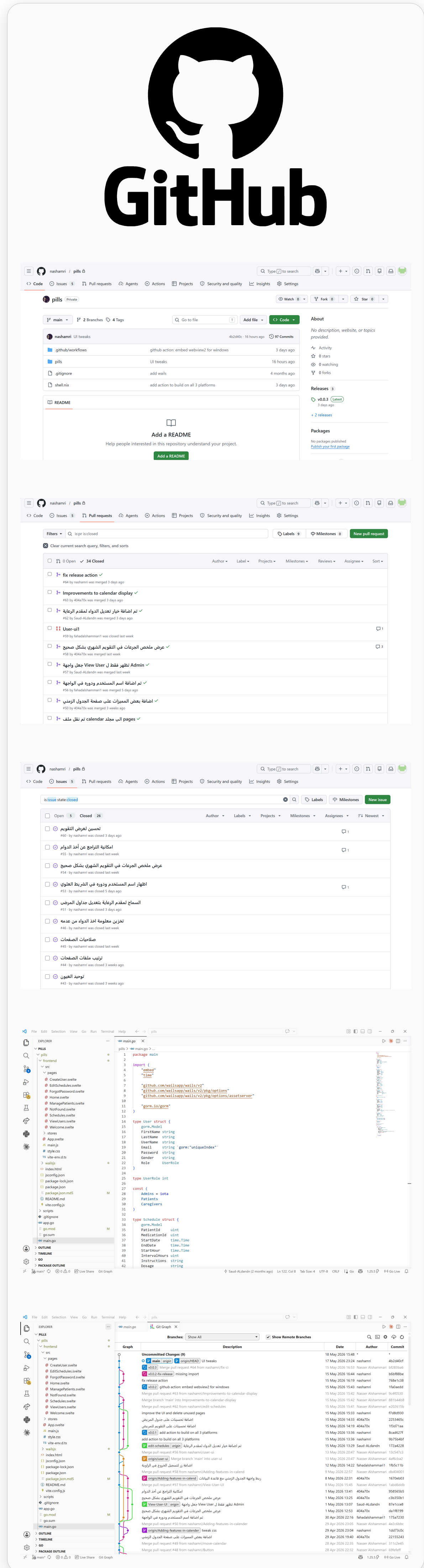
Application Interface



Technologies



Development Workflow



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